Focusly: Smart Task & Deadline Manager

Architecture: Browser → JavaScript → API → Java → Database

1. Browser

The student's phone or laptop.

- HTML + CSS (responsive, phone tab bar)
- Alarm at 9:00 AM & 8:00 PM
- Pop-up notifications

→ events, form data

2. JavaScript

`app.js`

- DOM + event handling
- Form validation
- `fetch()` + async/await
- Loading and error messages
- Search, sort, counter, calendar

⇄ HTTPS JSON + Bearer token

3. REST API

Spring Boot controllers

- POST /api/auth/register, /login
- GET /api/tasks, /stats, /focus
- POST /api/tasks
- PUT /api/tasks/{id}
- PATCH /api/tasks/{id}/status
- DELETE /api/tasks/{id}

→ method calls

4. Java

Business logic

- `TaskService`, `AuthService`
- `TaskValidator` (wrong input)
- `SmartScorer` (priority ranking)
- Exceptions → clean JSON errors
- BCrypt passwords, token hashes

⇄ JPA SQL

5. Database

Data survives restarts

- PostgreSQL on Render (live)
- MySQL or H2 file (local)
- Tables: users, tasks, auth_tokens

Hosting

Docker image on Render (web service) + Render PostgreSQL. HTTPS comes from Render.

Security

No secrets in code: database login comes from environment variables. Every request is checked by `AuthInterceptor`.

Phone apps (bonus)

Capacitor wraps the same website for Android and iOS and schedules the alarms on the phone.

Example journey: student taps “Add task” → JS validates the form → `fetch POST /api/tasks` → controller → `TaskValidator` + `TaskService` → saved in the database → JSON returned → JS redraws the list and the counters.